NAVLE Review

Swine Study Guide

A complete review of swine medicine,
organized by system and by age group.

Enteric Disease by Age · Respiratory Disease · Systemic & Septicemic
Neurologic · Musculoskeletal & Lameness · Dermatology
Reproduction · Parasitology · Toxicology & Nutrition
Reportable & Foreign Animal Disease

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HOW TO USE THIS GUIDE

Yellow boxes are board pearls and buzzword-to-diagnosis shortcuts. **Blue boxes** are look-alike discriminators. **Red boxes** flag genuine clinical dangers.

THE SINGLE MOST USEFUL SWINE EXAM STRATEGY

Swine medicine is AGE medicine. Nearly every porcine question gives you an age — neonate, nursery, grower-finisher, or sow — and the age alone narrows the differential to two or three diseases. The age tables in §1, §2, and Appendix B are the highest-yield pages in this guide.

R = reportable FAD = foreign animal disease Z = zoonotic

1 | Enteric Disease

1.1 The Piglet Diarrhea Algorithm

Agent	Age	Features
***E. coli* (ETEC)**	**1–5 days**	**ALKALINE pH diarrhea** — enteric colibacillosis. Secretory, watery, no blood. **Fimbrial adhesins: F4 (K88) affects NEONATES**; **F5 (K99)** causes ETEC in **1–5 day old** pigs; **F18 with SHIGA TOXIN causes EDEMA DISEASE**.
Rotavirus	**Nursing/weaned**	**ACIDIC pH diarrhea** in young pigs — malabsorptive from villous atrophy. **Prevent by vaccination.**
***C. perfringens* type C**	**1–5 days**	Produces **alpha PLUS BETA toxin → HEMORRHAGIC, necrotizing enteritis of the small intestine in 1–5 day old piglets.** Peracute; often found dead. **Prevent with PRE-FARROW VACCINATION of the sow. Treat with ANTITOXIN.**
***C. perfringens* type A**	Neonatal	**Alpha toxin only → MILD necrotic enteritis.** One of the top three causes of milk-scours in a 3-day-old piglet.
C. difficile	Neonatal	**MESOCOLON EDEMA in neonates** — the distinguishing lesion.
Cystoisospora suis	**7–11 days**	**The coccidian of pigs. High morbidity, LOW mortality** — causing **NON-hemorrhagic, pasty yellow diarrhea in piglets over 1 week old.** Fibrinonecrotic enteritis of the jejunum/ileum. **Treat with a coccidiostat such as PONAZURIL** (or toltrazuril).
Porcine astrovirus	**< 2 weeks**	**STAR-SHAPED appearance on electron microscopy** — hence the name. Diarrhea in piglets under 2 weeks.
Sapovirus / norovirus	**Older piglets**	**CUP-SHAPED DEPRESSIONS on EM** (calicivirus morphology — "calici" means cup). **Can be ZOONOTIC.**
Sapelovirus	Neonatal	A **PICORNAVIRUS** — **ICOSAHEDRAL**. Causes **enteritis plus cardiac or neurologic signs** in neonatal pigs.
PEDV (porcine epidemic diarrhea virus)	**All ages; worst in neonates**	A **CORONAVIRUS**. **High MORBIDITY with severe diarrhea in NAIVE herds** and **very high mortality in neonates** (approaching 100% under 7 days). **Management: FEEDBACK** — deliberately exposing sows to infected material to generate lactogenic immunity that protects the piglets.
TGE (transmissible gastroenteritis)	All ages	A coronavirus causing **vomiting and diarrhea**, closely related to PEDV. Endemic in the US — **neither zoonotic nor federally reportable.**

THE 3-DAY-OLD PIGLET WITH A FLACCID, THIN-WALLED JEJUNUM

The **top three causes of milk-scours in a 3-day-old piglet with a flaccid, thin-walled jejunum containing watery contents**:
1. ROTAVIRUS   2. *E. COLI*   3. *C. PERFRINGENS* TYPE A
The thin, flaccid wall tells you the process is **secretory or malabsorptive**, not necrotizing — which is what separates these from type C, where the intestine is thickened, dark, and hemorrhagic.

1.2 Enteric Disease of Older Pigs

Disease	Agent	Features & Treatment
Porcine proliferative enteropathy (ileitis)	***Lawsonia intracellularis***	**Affects the SMALL INTESTINE (ileum) of OLDER pigs** — grower-finishers. **"GARDEN HOSE GUT"** — a thickened, rigid, corrugated ileum. The **hemorrhagic form (PHE)** causes acute death with a blood-filled intestine in finishing pigs and gilts. **Note: the same organism causes "wet tail" (proliferative ileitis) in hamsters and proliferative enteropathy in horses.** Treat with tylosin, tiamulin, or tetracyclines.
Swine dysentery	***Brachyspira hyodysenteriae***	**MUCOHEMORRHAGIC DIARRHEA** — mucus and blood, in the **LARGE INTESTINE**. **Affects FEEDER-GROWER pigs at roughly 7–16 weeks.** **Treat with LINCOMYCIN** (or tiamulin).
Salmonellosis Z	***Salmonella*** Typhimurium and Choleraesuis	**Two lesions to name:** 1. **"BUTTON ULCERS"** — round, raised, concentric ulcers in the colon/cecum (also seen in classical swine fever) 2. **RECTAL STRICTURES** from healed ulceration and ischemia, specifically with *S.* Typhimurium **Two structures affected in swine: the intestine and the lung.** *S.* Choleraesuis is the septicemic/pneumonic form. **Prevention: vaccinate in the water or intranasally.** **Human GI disease comes from *S.* Typhimurium and *S.* Enteritidis.**
Edema disease	***E. coli*** **F18 with SHIGA TOXIN**	Post-weaning. Shiga toxin causes an **angiopathy** → **edema of the eyelids, forehead, stomach wall (gastric submucosa), and mesocolon**, plus **neurologic signs** (ataxia, paddling) and a characteristic **squeaky/hoarse squeal** from laryngeal edema. Sudden death in the best pigs. **Management: REDUCE WEANING STRESS** and abrupt feed changes.

2 | Respiratory Disease

Disease	Agent	Features & Management
Enzootic pneumonia	***Mycoplasma hyopneumoniae***	**A dry, non-productive "BARK" COUGH**, classically described in **naive sows** and growing pigs. **Also reduces weight gain and feed efficiency** — the economic cost is mostly subclinical. **Lesion: CRANIOVENTRAL CONSOLIDATION** — purple-gray, well-demarcated. **Treat with LINCOMYCIN** (or tylosin, tiamulin, tetracyclines) **and IMPROVE VENTILATION.** Destroys tracheal cilia, so it is the great enabler of secondary bacterial pneumonia.
PRRS (porcine reproductive and respiratory syndrome)	**ARTERIVIRUS** — an enveloped RNA virus	**Causes SEVERE NECROTIZING INTERSTITIAL PNEUMONIA and respiratory failure, and PREDISPOSES pigs to secondary infections** by destroying alveolar macrophages. **Reproductive form: MUMMIFIED fetuses and STILLBIRTHS** (the "SMEDI"-like picture), late-term abortion. **GILTS spread it** — introducing naive or infected replacement gilts is the classic breakdown. **Histopathology: interstitial pneumonia with type II pneumocyte hyperplasia and septal infiltration.** **A positive ELISA antibody test means EXPOSURE, not active infection or protection** — it cannot distinguish vaccination from field infection.
Actinobacillus pleuropneumonia (APP)	***Actinobacillus pleuropneumoniae***	**Affects GROWING pigs. OPEN-MOUTH BREATHING**, bloody froth at the nose, and **a sharp increase in mortality** — pigs die acutely with a fibrinohemorrhagic necrotizing pleuropneumonia and dorsocaudal lung lesions.
Atrophic rhinitis	***Bordetella bronchiseptica*** + **toxigenic *Pasteurella multocida***	**Both are GRAM-NEGATIVE COCCOBACILLI.** ***Bordetella* ALONE causes rhinitis. *Bordetella* COMBINED with toxigenic *Pasteurella* causes ATROPHIC RHINITIS with TURBINATE ATROPHY** — snout deviation, epistaxis, tear staining. That two-organism requirement is the exam point.
Swine influenza Z	Influenza A	**Seen in FALL and WINTER. ACUTE ONSET and RAPID RECOVERY** — high morbidity, low mortality, herd recovers in about a week. **Zoonotic and reverse-zoonotic.**
Mycoplasma hyorhinis	Mycoplasma	**SEPTICEMIA and POLYSEROSITIS in pigs <10 weeks** — see §3.
Glaesserella parasuis (Glässer's disease)	Gram-negative bacterium	**Septicemia, sudden death, and FIBRINOUS SEROSITIS** in **6–8 week old** pigs — see §3.
Ascarid larval migration	*Ascaris suum*	**VERMINOUS pneumonia** from larvae migrating through the lungs — see §8.

THE PORCINE RESPIRATORY DISEASE COMPLEX

Like BRDC in cattle, swine respiratory disease is usually **multifactorial**: a primary agent (***M. hyopneumoniae*, PRRS, influenza, PCV2**) damages defenses, and secondary bacteria (***Pasteurella*, *Glaesserella*, *Strep suis*, APP**) do the killing. Ventilation and stocking density are as much the treatment as any antibiotic.

3 | Systemic, Septicemic & Cardiac Disease

Disease	Agent	Features & Treatment
Streptococcus suis Z	Gram-positive coccus	**Found in the TONSILS and lungs; affects the JOINTS.** Seen in **WEANLING and GROWER** pigs. **Causes MENINGITIS** (paddling, opisthotonos), **arthritis**, and **VALVULAR ENDOCARDITIS**. **Three signs of valvular endocarditis: SEIZURES · PADDLING · FIBRIN IN THE CHEST.** **Zoonotic** — causes meningitis and deafness in abattoir workers.
Glaesserella (Haemophilus) parasuis Glässer's disease	Gram-negative	**Septicemia and SUDDEN DEATH in 6–8 WEEK OLD pigs.** **Sign: FIBRINOUS SEROSITIS** — fibrin on the pericardium, pleura, peritoneum, and joints. **Treat with PENICILLIN.**
Mycoplasma hyorhinis	Mycoplasma	**SEPTICEMIA in pigs UNDER 10 WEEKS** — arthritis with polyserositis and systemic disease → **failure to thrive**. **Associated with COLOSTRUM DEPRIVATION.**
Erysipelas Z	***Erysipelothrix rhusiopathiae*** — a gram-positive rod	**Three forms:** acute septicemia with **DIAMOND-SHAPED skin lesions**; **chronic ARTHRITIS causing LAMENESS**; and **VEGETATIVE VALVULAR ENDOCARDITIS**. **Treat with PENICILLIN** — highly susceptible. **Zoonotic: causes "erysipeloid" in fish handlers, butchers, and veterinarians.**
Mulberry heart disease	**Vitamin E and SELENIUM deficiency**	**TRANSMURAL HEMORRHAGES ON THE HEART** — a mottled, mulberry-like appearance. Sudden death in rapidly growing nursery pigs, typically the best in the pen. **Treat/prevent with VITAMIN E INJECTION** and dietary selenium. Related: **hepatosis dietetica** (liver) and **white muscle disease** (skeletal muscle) are the same deficiency in different tissues.
Mycoplasma suis (eperythrozoonosis)	Hemotropic mycoplasma	**An infectious HEMOLYTIC ANEMIA** — attaches to the red cell surface. **The classic transmission route: CONTAMINATED NEEDLES** (and biting insects). Icterus, pallor, poor doing. Treat with tetracyclines.
PCV2 / PMWS (post-weaning multisystemic wasting syndrome)	**PORCINE CIRCOVIRUS 2**	**Three findings: PALE KIDNEYS (with white foci) · ENLARGED LYMPH NODES · EDEMA / pleural effusion.** Progressive wasting in nursery and grower pigs, plus dermatitis-nephropathy syndrome. **Diagnosis: detect VIRAL ANTIGEN IN LYMPHOID TISSUE using IHC or IN SITU HYBRIDIZATION** — because the virus is ubiquitous, you must show it **in the lesion** to call it disease. Effective vaccines exist.
Leptospirosis Z	*Leptospira*	**ABORTION and SMALL LITTER SIZE.** Zoonotic.

4 | Neurologic Disease

Disease	Agent	Features
Pseudorabies (Aujeszky's disease) R	**HERPESVIRUS** (suid herpesvirus-1)	**Affects piglets < 7 days most severely** — near 100% mortality with CNS signs. **Lesion: NECROTIZING TONSILLITIS.** **FATAL TO CATS AND DOGS** that eat infected pork — the "mad itch," with frantic self-mutilation and rapid death. **Pigs are the only natural host; every other species dies.** **Eradicated from commercial US herds — reportable**; feral swine remain a reservoir.
Porcine polio (porcine enteroviral encephalomyelitis, Teschen/ Talfan disease)	**TESCHOVIRUS**	**Affects ALL AGES.** Four signs: **fever · lethargy · ataxia · a PARALYZED pig.** Progressive posterior paresis to tetraplegia.
Edema disease	*E. coli* F18 Shiga toxin	Neurologic signs from cerebrovascular angiopathy — ataxia, paddling, and the squeaky squeal. See §1.2.
Salt poisoning / water deprivation	Sodium ion toxicosis	**PIGS ARE THE MOST AFFECTED SPECIES.** **PATHOGNOMONIC LESION: PERIVASCULAR INFILTRATION OF EOSINOPHILS ("eosinophilic cuffing") in the brain.** **Mechanism:** high blood sodium → brain dehydration → the brain generates **idiogenic osmoles** → when water is suddenly restored, water rushes back in and the brain swells → **head pressing, star-gazing, blindness, seizures**, "dog-sitting" posture. **The critical management point: REINTRODUCE WATER SLOWLY AND IN SMALL AMOUNTS.** Giving free access to water is what kills them.
***Strep suis* meningitis**	*S. suis*	Weanlings; paddling and opisthotonos. See §3.
Sapelovirus	Picornavirus	Neonatal enteritis **plus cardiac or neurologic signs**.

5 | Musculoskeletal & Lameness

Condition	Agent	Features & Treatment
***Mycoplasma hyorhinis* arthritis**	Mycoplasma	**Pigs UNDER 10 WEEKS. Arthritis WITH polyserositis and SYSTEMIC disease** → failure to thrive. **Associated with colostrum deprivation.**
***Mycoplasma hyosynoviae* arthritis**	Mycoplasma	**Pigs 4–6 MONTHS old. MILD lameness LOCALIZED TO JOINTS**, with no systemic illness — think **chronic lameness** in a grower-finisher. **Associated with poor ventilation, high ammonia, and stress. RESISTANT TO PENICILLIN — treat with TYLOSIN or LINCOMYCIN.**
Septic arthritis	***Streptococcus***	In growing pigs, usually associated with **trauma or hematogenous seeding** (tail biting, navel infection, castration wounds).
Erysipelas arthritis	*E. rhusiopathiae*	Chronic proliferative arthritis and lameness; penicillin.
Porcine stress syndrome / malignant hyperthermia	**RYR1 mutation** (halothane gene)	Triggered by stress or halothane anesthesia. Uncontrolled calcium release from the sarcoplasmic reticulum → hyperthermia, muscle rigidity, acidosis, death. Produces **PSE (pale, soft, exudative) pork**. **Treat and prevent with DANTROLENE — a RYANODINE RECEPTOR ANTAGONIST** that reduces muscle contraction by blocking calcium release.

***M. HYORHINIS* VS. *M. HYOSYNOVIAE* — THE TWO MYCOPLASMA ARTHRITIDES**

hyoRHINIS = YOUNGER (<10 weeks) = SYSTEMIC (septicemia + polyserositis + arthritis) = colostrum-deprived.
hyoSYNOVIAE = OLDER (4–6 months) = JOINTS ONLY = mild chronic lameness = ventilation and stress.
Mnemonic: **syn**oviae stays in the **syn**ovium.

6 | Dermatology

Condition	Agent	Features & Treatment
Greasy pig disease (exudative epidermitis)	***Staphylococcus hyicus***	**BROWN, greasy exudate covering pigs UNDER 6 WEEKS of age.** Generalized, non-pruritic, and often fatal from dehydration and protein loss in severe cases. Follows skin abrasion (fighting, rough floors, mange).
Sarcoptic mange Z	*Sarcoptes scabiei* var. *suis*	The most important ectoparasite of pigs — intense pruritus, ear crusts, poor growth. **Treat with PERMETHRIN and INJECTABLE IVERMECTIN.**
"Dippity pig" (erythema multiforme)	Immune-mediated	**A TYPE III HYPERSENSITIVITY causing VASCULITIS. Lesion: CONCENTRIC, IRIS-LIKE RINGS of erythema** — target lesions, classically over the back, with acute pain and dipping of the back. Self-limiting.
Pityriasis rosea	Idiopathic/hereditary	**RINGWORM-LIKE (annular) lesions in pigs OVER 2 MONTHS of age.** Self-limiting and non-pruritic — the important point is that it is **not** ringworm and needs no treatment.
Zinc-responsive dermatitis	Zinc deficiency	**PARAKERATOSIS** — thick, crusted, scaly plaques. Excess dietary calcium binds zinc and precipitates it. Supplement zinc.
Epitheliogenesis imperfecta	Congenital	**Congenital absence or defect of the epidermis**, typically on the **legs or over the hooves** — well-demarcated areas of missing skin present at birth.
Seneca Valley virus	Senecavirus A	**Causes vesicles INDISTINGUISHABLE from FMD — but is NOT reportable itself. This is the great practical trap in swine medicine:** any vesicular lesion in a pig must be treated as a possible foreign animal disease and **investigated** until FMD, VS, swine vesicular disease, and vesicular exanthema are ruled out.

7 | Reproduction

Topic	Detail
Estrous cycle	**Year-round polyestrus; 21-day cycle. Estrus lasts 40–60 hours** overall — **36–48 hours in GILTS** and **48–72 hours in SOWS**.
Gestation	**114 days — "3 months, 3 weeks, 3 days."**
Weaning to estrus	**Multiparous sows return to estrus 3–7 days after weaning.**
Pregnancy diagnosis	**Reliable after 30 days** by ultrasound.
Fetal calcification	Visible radiographically at about **35 days** — useful for aging mummified fetuses.
Porcine parvovirus	**Two effects: IRREGULAR RETURN TO ESTRUS and DECREASED LITTER SIZE.** **The animal most affected is the healthy FIRST-PARITY GILT**, which is naive — she produces **dead and mummified fetuses**. Part of the **SMEDI** complex (Stillbirth, Mummification, Embryonic Death, Infertility). **Vaccinate gilts before breeding** — this is why gilt acclimation exists.
PRRS	**Mummified fetuses and stillbirths**, late-term abortion. **Gilts spread it.**
Leptospirosis	**Abortion and small litter size.**
Endometritis	Two causes: ***Staphylococcus hyicus*** and ***E. coli***.
Rectal prolapse	Two causes: **COUGHING** and **MYCOTOXINS — specifically ZEARALENONE** (its estrogenic effect causes tenesmus and perineal edema).
Zearalenone	**PIGS are the sensitive species** because the metabolite is **ESTROGENIC**. Causes **HYPERESTROGENISM and PSEUDOPREGNANCY**, vulvar swelling in prepubertal gilts, and rectal/vaginal prolapse. See §9.

8 | Parasitology

Parasite	Location	Disease & Treatment
Ascaris suum	**Adults in the SMALL INTESTINE; larvae migrate through the LIVER and LUNGS**	**"MILK SPOTS" ON THE LIVER** — white fibrotic tracts from larval migration; the classic slaughterhouse condemnation lesion. **Larvae travel to the LUNGS → VERMINOUS PNEUMONIA** with coughing and mild diarrhea. **Treat with FENBENDAZOLE.** Piperazine also works but is a paralytic — in a very heavy burden, rapid paralysis of a large worm mass risks obstruction. **Roundworms in pigs are found mostly in the small intestine.**
Trichuris suis (porcine whipworm)	**CECUM and COLON**	**Commonly found in pigs UNDER 6 MONTHS old, causing diarrhea** — mucohemorrhagic in heavy burdens, and a mimic of swine dysentery. **Treat with FENBENDAZOLE.**
Stephanurus dentatus (swine kidney worm)	**LIVER and KIDNEY** (perirenal fat and ureters)	Long prepatent period; liver condemnation. **Control is entirely by MANAGEMENT — three steps:** 1. **Replace older boars with young boars from a clean herd** 2. **Breed only gilts to those young boars** 3. **Sell the gilts and young boars to slaughter after weaning their first litter** The logic: the parasite's long life cycle means removing all animals before they become patent breaks transmission. **A "gilt-only" facility eliminates it.**
Cystoisospora suis	Small intestine	Coccidiosis at **7–11 days**; non-hemorrhagic diarrhea; **ponazuril**.
Trichinella spiralis Z	**Muscle (encysted larvae)**	**A ROUNDWORM (nematode). In the pig: fever, MYOSITIS, and edema.** **Treatment is very difficult** — the answer is to **remove the pig from the food chain**, or an azole. **The public health answer: COOK PORK TO 160°F.** Freezing also kills most species. **Zoonotic** — human trichinellosis from undercooked pork or wild game.
Sarcoptic mange	Skin	**Permethrin and injectable ivermectin.**

9 | Toxicology & Nutritional Disease

Toxin / Deficiency	Source & Mechanism	Signs & Management
Zearalenone	A ***Fusarium*** **mycotoxin** in moldy corn. **PIGS are sensitive because the metabolite is ESTROGENIC** (zearalenol binds estrogen receptors).	**Two effects: HYPERESTROGENISM and PSEUDOPREGNANCY.** Vulvar swelling and reddening in prepubertal gilts, vaginal and **rectal prolapse**, anestrus, and reduced litter size.
Fumonisin	***Fusarium*** in moldy corn. (In horses the same toxin causes leukoencephalomalacia.)	**Three effects in pigs: HYDROTHORAX · PORCINE PULMONARY EDEMA · respiratory signs.** Also hepatotoxicity.
Aflatoxin	***Aspergillus flavus*-contaminated CORN.**	**Hepatotoxic — three signs: ASCITES · ICTERUS · ANOREXIA.** Also immunosuppression and reduced growth.
Salt poisoning / water deprivation	**PIGS ARE THE MOST AFFECTED SPECIES.**	**Pathognomonic: PERIVASCULAR EOSINOPHILIC CUFFING** in the brain. Head pressing, blindness, seizures. **Reintroduce water SLOWLY.**
Iron toxicity	Iron dextran injection in a selenium/ vitamin E-deficient piglet.	**Three effects: ACUTE MUSCLE DAMAGE · HYPERKALEMIA · DEATH.** Sudden death within hours of injection — a genuine and avoidable clinical disaster. Ensure sow selenium/vitamin E status before routine iron dosing.
Vitamin E / selenium deficiency	Dietary.	**Three syndromes: MULBERRY HEART DISEASE** (transmural cardiac hemorrhage), **hepatosis dietetica**, and **white muscle disease**. **Treat with vitamin E injection.**
Cocklebur	**PIGS are the most affected species**; the toxic stage is the **two-leaf seedling**.	**Acute hepatic necrosis** with hypoglycemia, depression, and death.
Copper	**PIGS are RESISTANT to copper toxicity** relative to sheep — copper is fed as a growth promoter.	The exam point is the contrast: **sheep are exquisitely sensitive; pigs are not.** (Toxicity still occurs at very high dietary levels.)
Gossypol	Cotton seed.	Cardiotoxic. **Inactivated by IRON.**

Appendix A | Reportable & Foreign Animal Diseases

VESICLES IN A PIG ARE A FOREIGN ANIMAL DISEASE UNTIL PROVEN OTHERWISE

FMD, vesicular stomatitis, swine vesicular disease, and vesicular exanthema of swine are clinically indistinguishable — and so is **Seneca Valley virus**, which is not reportable but looks identical. **Any vesicular lesion in a pig must be reported and investigated.** Notify the State Animal Health Official or USDA Area Veterinarian in Charge.

Disease	Agent	Recognition	Action
Classical swine fever (hog cholera) FAD	**Pestivirus**	High fever, skin hemorrhage and cyanosis of the extremities, **button ulcers**, "huddling," neurologic signs, high mortality. **Eradicated from the US in 1978.**	**DEPOPULATE THE HERD.** Report immediately.
African swine fever FAD	Asfivirus	Clinically similar to CSF with near-100% mortality; no vaccine.	Report; depopulate.
Foot and Mouth Disease FAD	Aphthovirus	Vesicles on the snout, coronary bands, and teats. **Pigs are efficient amplifiers.**	Report immediately.
Vesicular stomatitis R Z	Vesiculovirus	**Affects PIGS, HORSES, and SHEEP.** Indistinguishable from FMD in the pig.	Report.
Pseudorabies R	Herpesvirus	Necrotizing tonsillitis; CNS disease in piglets; **fatal "mad itch" in dogs and cats.**	Report; eradicated from commercial herds, feral swine reservoir.
Swine influenza Z	Influenza A	Acute onset, rapid recovery, fall/winter.	Zoonotic and reverse-zoonotic; monitor for novel reassortants.
Trichinella Z	Nematode	Muscle cysts; human disease from undercooked pork.	**Cook pork to 160°F.**
Streptococcus suis Z	Gram-positive coccus	Meningitis and deafness in abattoir and farm workers.	Occupational hygiene.
Erysipelas Z	*E. rhusiopathiae*	"Erysipeloid" — painful hand lesions in handlers, butchers, and fish workers.	Gloves; penicillin.
Brucellosis R Z	*Brucella suis*	Abortion, orchitis; feral swine reservoir.	Report; test and slaughter.
Salmonella Z	*S.* Typhimurium, Enteritidis	Human GI disease.	Food safety controls.

Appendix B | Ages, Numbers & Timelines

B.1 Disease by Age — the master table

Age	Think of
< 7 days	**Pseudorabies** (near 100% mortality) · ***E. coli* F4/F5** · ***C. perfringens* type C** (1–5 d) · ***C. difficile*** (mesocolon edema) · **porcine astrovirus** · **PEDV** · **sapelovirus**
7–11 days	***Cystoisospora suis*** coccidiosis
< 6 weeks	**Greasy pig disease (*Staph hyicus*)**
6–8 weeks	***Glaesserella parasuis*** — septicemia, sudden death, fibrinous serositis
< 10 weeks	***Mycoplasma hyorhinis*** — systemic arthritis with polyserositis
Post-weaning	**Edema disease (F18 Shiga toxin)** · **PCV2 / PMWS**
7–16 weeks	**Swine dysentery (*Brachyspira*)** — mucohemorrhagic diarrhea
< 6 months	***Trichuris suis*** — whipworm diarrhea
4–6 months	***Mycoplasma hyosynoviae*** — mild chronic joint lameness
Grower-finisher	***Lawsonia* (garden hose gut)** · **APP** · ***Strep suis*** · **erysipelas**
First-parity gilt	**Porcine parvovirus** — mummies and small litters · **PRRS**
Over 2 months	**Pityriasis rosea** (ringworm-like, self-limiting)

B.2 Numbers

Number	Meaning
Gestation 114 days	"3 months, 3 weeks, 3 days."
Cycle 21 days; year-round polyestrus	Estrus 40–60 h overall.
Gilt estrus 36–48 h; sow estrus 48–72 h	Sows stand longer than gilts.
Weaning to estrus 3–7 days	Multiparous sows.
Pregnancy diagnosis > 30 days	Fetal calcification visible at ~35 days.
Pig teeth: 44	The most of any domestic species.
Cook pork to 160°F	*Trichinella* control.